

FAB Group Project

Weekly Module 4: Knowledge Representation & Reasoning

Objective | Define: (1) a concept **ontology** out of your **variables** and their **metadata**, and extend this ontology beyond the course notes to a large data set; (2) extract **knowledge representations** (e.g., “claims”) based on this ontology; (3) identify a set of questions for **reasoning** about this knowledge.

Resources | (1) PDFs of Course Notes covered so far (Credit Assignment, Info. Proc., Search, Evolutionary Strategies; Representation; Knowledge Representation and Reasoning); (2) A set of operators, heuristics, prompts, etc.; (3) (4) The definition of your goal state and constraint networks from previous sessions; (5) A classwide list of concepts, metadata, and constraints (see KLMS); (6) A large text dataset (around 500 academic papers) on AI and business.

Current gap | Current AI tools support prompting based on analyzing statistical patterns among word fragments (called tokens) in text. This method is very powerful, but often lacks precision. To develop an AI system that enables generally reasoning about AI ideas, it will help to represent text more precisely, as a formal ontology and in terms of specific knowledge claims and their contexts. Doing so will enable fine-grained, flexible, and precise AI outputs that should enable rich reasoning compared to just generating summaries in an ad hoc fashion.

Recommended Steps | I recommend proceeding in the following order:

0. Make sure you understand the concepts of common sense, concepts, contexts, and elaboration tolerance from the course notes on knowledge representation and reasoning.
1. Refer to your constraint network that you defined for the last session (your variables, values, and their constraints), and the classwide list of concepts, metadata, and constraints..
2. Discuss how a student might typically wish to **reason** about the knowledge in the course notes. What **questions** might a student want to ask, either of the course notes and/or the overall dataset? Write a set of these questions down.
3. Using the classwide list, organize the concepts into an ontology. This will require: a) defining your set of concepts; b) defining their relations in terms of various constraints and metadata (definitions, level of abstraction, author, etc.); c) cleaning up (normalizing) the ontology by removing duplicates and accounting for synonyms.
4. Using an LLM, construct a **knowledge base** by extracting knowledge claims among one or more of these concepts (causal constraints, correlative constraints, contradictions, conditions, etc.), and their contexts (**authors, topics, papers, etc.**) from the text of the course notes only. Then, input your knowledge base into an LLM, and ask reasoning questions from (2.) above.
6. Take a **screenshot** or photo of your constraint network, write a **short paragraph** describing what you have done, save as a **PDF**, and **upload on KLMS**.